

matrixH.C Options Summary

Signature:

```
matrixH(TString trainAO2D = "AO2D.root", bool useAO2DTraining = false, TString dataAO2D = "AO2Ddata.root", TString truthAO2D = "AO2Dtruth.root", int useAO2Data = 2, int priorMode = 0, bool subtractFakeFeed = true, double centMin = 0.0, double centMax = 100.0, int sameBCPolicy = 0)
```

Main Switches

trainAO2D	MC training file. Used for the response matrix when useAO2DTraining = true.
useAO2DTraining	false: generate toy response. true: read response/training from the MC AO2D table O2nmpmcchargedtabl.
dataAO2D	Reco-data-like input. For real data this is read from O2npcollisiontabl plus O2nprecochargedca.
truthAO2D	Truth input for independent-MC closure when useAO2Data = 3.
useAO2Data	0: generated pseudo-data. 1: real reco data. 2: RecoTrain/TruthTrain same-sample closure. 3: independent MC2 reco-table plus MC2 truth closure. 4: MC2 reco and truth both read from O2nmpmcchargedtabl.
priorMode	0: flat positive prior. 1: TruthTrain fixed-point/debug prior. 2: RecoTrain-shaped prior. 3: Poisson-randomized TruthTrain prior. 4: Poisson-randomized flat positive prior.
subtractFakeFeed	true: subtract fake/feed-in estimate from reco input before unfolding. false: bypass fake/feed subtraction.
centMin/centMax	FT0M centrality percentile interval applied only when reading data-like reco collision tables O2npcollisiontabl. Default 0-100 keeps all collisions.
sameBCPolicy	Applied only to data-like reco collision tables. 0: use fCentFT0M and no pileup cut. 1: require stored fNoSameBunchPileup and use fCentFT0MNoPileup. Option 2 was removed.

Useful Output Histograms

hUnfold	Final unfolded 2D result in pt and multiplicity.
hUnfoldPtProj	pt projection of hUnfold.
hUnfoldMultProj	track-weighted multiplicity projection of hUnfold.
hUnfoldMultEventProj	hUnfoldMultProj divided bin-by-bin by multiplicity M; approximate event-style multiplicity view.
hTruthDataMultEventProj	hTruthDataMultProj divided bin-by-bin by multiplicity M; closure reference for hUnfoldMultEventProj.
hTruthTrainMultEventProj	hTruthTrainMultProj divided bin-by-bin by multiplicity M; same-sample/debug reference.

hRatioPt	$\text{hUnfoldPtProj} / \text{hTruthDataPtProj}$.
hRatioMult	$\text{hUnfoldMultProj} / \text{hTruthDataMultProj}$.
hRatioMultEvent	$\text{hUnfoldMultEventProj} / \text{hTruthDataMultEventProj}$.
QA/	Output directory containing fake/feed and applied-purity QA plots.
QA/hFakeFeedFractionPtProj	(fake + feed-in) / all reco training versus pt.
QA/hFakeFeedFractionMultProj	(fake + feed-in) / all reco training versus multiplicity.
QA/hFakeFeedOverSignalPtProj	(fake + feed-in) / matched signal reco training versus pt.
QA/hFakeFeedOverSignalMultProj	(fake + feed-in) / matched signal reco training versus multiplicity.
QA/hRecoDataPurityPtProj	$\text{hRecoDataUnfoldPtProj} / \text{hRecoDataPtProj}$; the actual fake/feed purity applied to data versus pt.
QA/hRecoDataPurityMultProj	$\text{hRecoDataUnfoldMultProj} / \text{hRecoDataMultProj}$; the actual fake/feed purity applied to data versus multiplicity.

Closure / Data Modes

1. Toy Response + Toy Data

Switches	useAO2DTraining = false, useAO2Data = 0
Meaning	Training response is generated by the toy model. Pseudo-data is generated independently by the same toy model.
Compare	hUnfold to hTruthData. Projections: hUnfoldPtProj to hTruthDataPtProj, hUnfoldMultProj to hTruthDataMultProj, hUnfoldMultEventProj to hTruthDataMultEventProj. Ratios: hRatioPt, hRatioMult, hRatioMultEvent.
Example	<code>matrixH("A02D.root", false, "A02Ddata.root", "A02Dtruth.root", 0, 0, true)</code>

Closure / Data Modes

2. Same-Sample Closure / Debug

Switches	useAO2Data = 2. useAO2DTraining can be false for toy response or true for AO2D MC response.
Meaning	RecoData is replaced with RecoTrain and TruthData with TruthTrain.
Compare	hUnfold to hTruthTrain. Projections: hUnfoldPtProj to hTruthTrainPtProj, hUnfoldMultProj to hTruthTrainMultProj, hUnfoldMultEventProj to hTruthTrainMultEventProj. With priorMode = 1 this is the fixed-point/debug check.
Example	<code>matrixH("A02D.root", true, "A02Ddata.root", "A02Dtruth.root", 2, 1, false)</code>

Closure / Data Modes

3. MC Response + Real Data

Switches	useAO2DTraining = true, useAO2Data = 1
Meaning	trainAO2D provides the MC response/training. dataAO2D provides real reco data. truthAO2D is not used. centMin/centMax select the FT0M centrality percentile range in O2npcollisiontabl.
Pileup/centrality	sameBCPolicy = 0 uses fCentFT0M without a pileup cut. sameBCPolicy = 1 requires fNoSameBunchPileup and uses fCentFT0MNoPileup. A NoPileupCentrality.root-style file must be a full AO2D copy with O2nprecochargedca plus enriched O2npcollisiontabl.
Compare	For real data there is no truth closure target. Inspect hUnfold, hUnfoldPtProj, hUnfoldMultProj, and hUnfoldMultEventProj. Compare to external expectations or systematic variations, not to hTruthData.
QA	Check QA/hRecoDataPurityMultProj and QA/hFakeFeedFractionMultProj to understand the fake/feed correction size. A flat dN/deta deficit can come from a flat purity correction.
Example	<code>matrixH("AO2D.root", true, "NoPileupCentrality.root", "AO2Dtruth.root", 1, 0, true, 0, 10, 1)</code>

Closure / Data Modes

4. MC1 Response + MC2 Reco + MC2 Truth

Switches	useAO2DTraining = true, useAO2Data = 3
Meaning	trainAO2D = MC1 response/training. dataAO2D = MC2 reco-data-like input. truthAO2D = MC2 truth. centMin/centMax and sameBCPolicy select the reco/data-like collisions only.
Pileup/centrality	sameBCPolicy = 1 uses stored fNoSameBunchPileup and fCentFT0MNoPileup. There is no internal run/globalBC/NumContrib pileup calculation in matrixH.C anymore.
Compare	hUnfold to hTruthData. Projections: hUnfoldPtProj to hTruthDataPtProj, hUnfoldMultProj to hTruthDataMultProj, hUnfoldMultEventProj to hTruthDataMultEventProj. Ratios: hRatioPt, hRatioMult, hRatioMultEvent.
Caveat	With a non-inclusive centrality range, hTruthData remains inclusive unless the truth table also contains centrality. Use this mode for centrality closure only after adding a centrality-selected truth target.
Example	<code>matrixH("A02D_MC1.root", true, "A02D_MC2_data.root", "A02D_MC2_truth.root", 3, 3, true, 0, 10, 1)</code>

Closure / Data Modes

5. MC1 Response + MC2 NPMC Reco + MC2 NPMC Truth

Switches	useAO2DTraining = true, useAO2Data = 4
Meaning	trainAO2D = MC1 response/training. dataAO2D = MC2 O2nPMCchargedtab1; reco pseudo-data comes from fPtRec/fMultNTracksNP and truth target from fPtGen/fMultGen. truthAO2D is not used.
Compare	hUnfold to hTruthData. Projections: hUnfoldPtProj to hTruthDataPtProj, hUnfoldMultProj to hTruthDataMultProj, hUnfoldMultEventProj to hTruthDataMultEventProj. Ratios: hRatioPt, hRatioMult, hRatioMultEvent.
Caveat	This mode does not fill true collision-level hMultRecoData, because O2nPMCchargedtab1 is row-based rather than a collision table. centMin/centMax and sameBCPolicy are not applied in this mode because the row table has no centrality or fNoSameBunchPileup/fCentFTOMNoPileup collision columns.
Example	<code>matrixH("A02D_MC1.root", true, "A02D_MC2.root", "A02Dtruth.root", 4, 4, true)</code>

Postprocessing / Cross-Checks

dN/deta vs Centrality

Macro	plotDnDetaVsCentrality.C reads the ten centrality output ROOT files and plots dNch/deta versus FT0M centrality.
Normalization	$dN/deta = \text{Integral}(h\text{UnfoldMultProj}) / \text{Integral}(h\text{UnfoldMultEventProj}) / \text{etaWidth}$. It is not divided by centrality-bin width.
Default etaWidth	1.6 for $ \text{eta} < 0.8$.
Reference	The macro overlays the average of Table 6.1 periods LHC22o, LHC24aI, LHC24ao, and LHC24af, and writes hRatioToTable6p1Average.
Example	<code>plotDnDetaVsCentrality(".", "DnDetaVsCentrality.root", 1.6)</code>